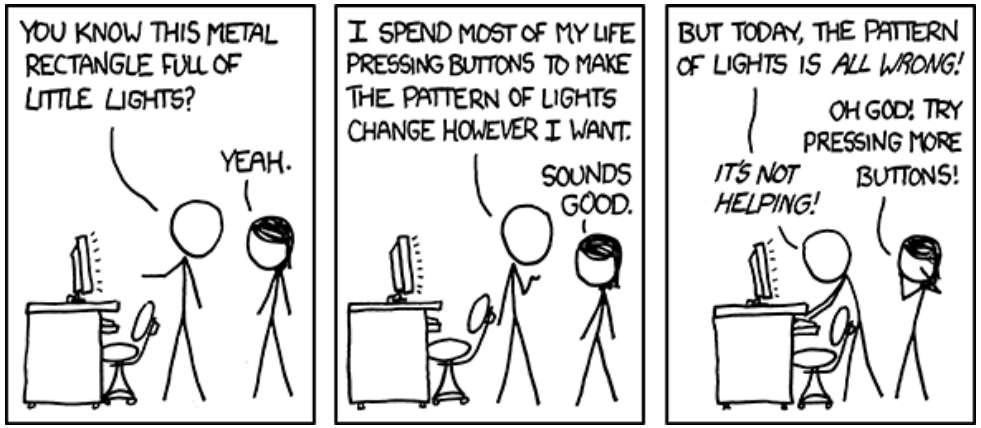
Admin



Assign 1 due tomorrow 5pm

Nab that on-time bonus!

Pre-lab for lab2

Read gcc/make guides
Watch video of Ben Eater's mad breadboard skills
Read about 7-segment display

Today: C functions

Implementation of C function calls

Coordinate of register use, runtime stack

From C to assembly (cont'd)

Previously

- C variable ⇒ registers
- C arithmetic/logical expression ⇒ ALU instructions
- C control flow ⇒ branch instructions
- C pointer ⇒ memory address
- Read/write memory ⇒ load/store instructions
- Array/struct data layout ⇒ address arithmetic

Today

- Assembly implementation of function call/return, parameters, return value
- Conventions on register use, ABI
- Runtime stack, local variables

```
loop:

    sw      a1,0x40(a0)
    lui     a2,0x3f00
delay:
    addi    a2,a2,-1
    bne     a2,zero,delay

    sw      zero,0x40(a0)
    lui     a2,0x3f00
delay2:
    addi    a2,a2,-1
    bne     a2,zero,delay2

    j       loop
```

How to unify repeated code? …

```
loop:

  sw    a1,0x40(a0)
  j     pause

  sw    zero,0x40(a0)
  j     pause

  j     loop
```

```
pause:
  lui     a2,0x3f00
delay:
  addi    a2,a2,-1
  bne     a2,zero,delay

// but... where to go now?
```

RISC-V Instruction-Set

Erik Engheim <erik.engheim@ma.com>

Arithmetic Operation

Mnemonic	Instruction	Type	Description
ADD rd, rs1, rs2	Add	R	rd ← rs1 + rs2
SUB rd, rs1, rs2	Subtract	R	rd ← rs1 - rs2
ADDI rd, rs1, imm12	Add immediate	I	rd ← rs1 + imm12
SLT rd, rs1, rs2	Set less than	R	rd ← rs1 < rs2 ? 1 : 0
SLTI rd, rs1, imm12	Set less than immediate	I	rd ← rs1 < imm12 ? 1 : 0
SLTU rd, rs1, rs2	Set less than unsigned	R	rd ← rs1 < rs2 ? 1 : 0
SLTIU rd, rs1, imm12	Set less than immediate unsigned	I	rd ← rs1 < imm12 ? 1 : 0
LUI rd, imm20	Load upper immediate	U	rd ← imm20 << 12
AUIP rd, imm20	Add upper immediate to PC	U	rd ← PC + imm20 << 12
XORI rd, rs1, imm12	XOR immediate	I	rd ← rs1 ^ imm12
SLL rd, rs1, rs2	Shift left logical	R	rd ← rs1 << rs2
SRL rd, rs1, rs2	Shift right logical	R	rd ← rs1 >> rs2
SRA rd, rs1, rs2	Shift right arithmetic	R	rd ← rs1 >> rs2
SLLI rd, rs1, shamt	Shift left logical immediate	I	rd ← rs1 << shamt
SRLI rd, rs1, shamt	Shift right logical imm.	I	rd ← rs1 >> shamt
SRAI rd, rs1, shamt	Shift right arithmetic immediate	I	rd ← rs1 >> shamt

Load / Store Operations

Mnemonic	Instruction	Type	Description
LD rd, imm12(rs1)	Load doubleword	I	rd ← mem[rs1 + imm12]
LW rd, imm12(rs1)	Load word	I	rd ← mem[rs1 + imm12]
LH rd, imm12(rs1)	Load halfword	I	rd ← mem[rs1 + imm12]
LB rd, imm12(rs1)	Load byte	I	rd ← mem[rs1 + imm12]
LWU rd, imm12(rs1)	Load word unsigned	I	rd ← mem[rs1 + imm12]
LHU rd, imm12(rs1)	Load halfword unsigned	I	rd ← mem[rs1 + imm12]
LBU rd, imm12(rs1)	Load byte unsigned	I	rd ← mem[rs1 + imm12]
SD rs2, imm12(rs1)	Store doubleword	S	rs2 → mem[rs1 + imm12]
SW rs2, imm12(rs1)	Store word	S	rs2(31:0) → mem[rs1 + imm12]
SH rs2, imm12(rs1)	Store halfword	S	rs2(15:0) → mem[rs1 + imm12]
SB rs2, imm12(rs1)	Store byte	S	rs2(7:0) → mem[rs1 + imm12]
BLT rs1, rs2, imm12	Branch less than	SB	if rs1 < rs2 PC ← PC + imm12
BLTU rs1, rs2, imm12	Branch less than unsigned	SB	if rs1 < rs2 PC ← PC + imm12 << 1
JAL rd, imm20	Jump and link	UJ	rd ← PC + 4 PC ← PC + imm20
JALR rd, imm12(rs1)	Jump and link register	I	rd ← PC + 4 PC ← rs1 + imm12

Pseudo Instructions

Mnemonic	Instruction	Base instruction(s)
LI rd, imm12	Load immediate (near)	ADDI rd, zero, imm12
LI rd, imm	Load immediate (far)	LUI rd, imm[31:12] ADDI rd, rd, imm[11:0]
LA rd, sym	Load address (far)	AUIPC rd, sym[31:12] ADDI rd, rd, sym[11:0]
MV rd, rs	Copy register	ADDI rd, rs, 0
NOT rd, rs	One's complement	XORI rd, rs, -1
NEG rd, rs	Two's complement	SUB rd, zero, rs
BGT rs1, rs2, offset	Branch if rs1 > rs2	BLT rs2, rs1, offset
BLE rs1, rs2, offset	Branch if rs1 ≤ rs2	BGE rs2, rs1, offset
BGTU rs1, rs2, offset	Branch if rs1 > rs2 (unsigned)	BLTU rs2, rs1, offset
BLEU rs1, rs2, offset	Branch if rs1 ≤ rs2 (unsigned)	BGEU rs2, rs1, offset
CALL offset	Call subroutine (far)	AUIPC ra, offset[31:12] JALR ra, ra, offset[11:0]
RET	Return from subroutine	JALR zero, 0(ra)
NOP	No operation	ADDI zero, zero, 0

JAL rd, imm20	Jump and link	UJ	rd ← pc + 4 pc ← pc + imm20
JALR rd, imm12(rs1)	Jump and link register	I	rd ← pc + 4 pc ← rs1 + imm12

32-bit instruction format

	31 30 29 28 27 26 25	24 23 22 21 20	19 18 17 16 15	14 13 12	11 10 9 8 7	6 5 4 3 2 1 0
R	func	rs2	rs1	func	rd	opcode
I	immediate		rs1	func	rd	opcode
SB	immediate	rs2	rs1	func	immediate	opcode
UJ	immediate				rd	opcode

Register File

r0	r1	r2	r3
r4	r5	r6	r7
r8	r9	r10	r11
r12	r13	r14	r15
r16	r17	r18	r19
r20	r21	r22	r23
r24	r25	r26	r27
r28	r29	r30	r31

Register Aliases

zero	ra	sp	gp
tp	t0	t1	t2
s0/fp	s1	a0	a1
a2	a3	a4	a5
a6	a7	s2	s3
s4	s5	s6	s7
s8	s9	s10	s11
t3	t4	t5	t6

ra - return address
sp - stack pointer
gp - global pointer
tp - thread pointer

t0 - t6 - Temporary registers
s0 - s11 - Saved by callee
a0 - 17 - Function arguments
a0 - a1 - Return value(s)

```
loop:

    sw    a1,0x40(a0)
    jal   ra,pause

    sw    zero,0x40(a0)
    jal   ra,pause

    j     loop
```

Right before jump, "write down" address of next instruction, save that address in register `ra`

Effectively `ra=pc+4`

```
pause:
  lui     a2,0x3f00
delay:
  addi    a2,a2,-1
  bne     a2,zero,delay
  jr      ra
```

Return at instruction whose address saved in register `ra`

Effectively `pc=ra`

pseudo instruction expands to `jalr zero,0(ra)`

```
loop:

  sw    a1,0x40(a0)
  lui   a2,0x3f00
  jal   ra,pause


  sw    zero,0x40(a0)
  lui   a2,0x3f00
  jal   ra,pause

  j     loop
```

How to communicate arguments to function?

```
pause:
delay:
  addi    a2,a2,-1
  bne     a2,zero,delay
  jr      ra
```

Add'l RISC-V instructions

jal **jump and link**

Save pc+4 to rd, jump target (target val immediate)

```
jal rd,imm                    // rd = pc+4, pc = pc+imm
```

jalr **jump and link register**

Save pc+4 to rd, jump target (target val in register, +offset)

```
jalr rd,imm(rs)          // rd = pc+4, pc = rs+imm
```

Pseudo-instruction conveniences

```
call fn    ->    jal ra,fn
jr reg     ->    jalr zero,0(reg)
ret        ->    jalr zero,0(ra)
```

Anatomy of C function call

```
int factorial(int n)
{
   int result = 1;
   for (int i = n; i > 1; i--)
      result *= i;
   return result;
}
```

Control flow: call and resume

Mechanism to pass arguments, return value

Shared used of registers, scratch space

Storage for local variables

Complication: nested function calls, recursion

Application binary interface

ABI specifies low-level machine interface and conventions that must be followed

- Mechanism for call/return
- How parameters passed
- How return value communicated
- Register use (ownership/preservation)
- Stack management (up/down, alignment)

Functions defined in separate modules/libraries must adhere to same ABI in order to successfully interoperate

Mechanics of call/return

Pass up to 8 args in registers **a0-a7**

call (**jal**) saves pc+4 to **ra**, jump to target

```
li a0,100
li a1,7
call sum
```

```
sum(100, 7);
```

```
int sum(int a, int b) {
    return a + b;
}
```

Return value put into **a0**

ret (**jalr**) jumps back to **ra** (i.e. resume previous)

```
add a0,a0,a1
ret
```

Caller and Callee

caller: function doing the calling

callee: function being called

```
void main(void) {
   gpio_init();
   timer_init();
   config_clock();
   clock_run();
   ...
}

void config_clock(void) {
    gpio_set_output(…);
    ...
}

void gpio_set_output(gpio_id_t) {
  ...
```

`main` is <u>caller</u> of **`config_clock`**

`config_clock` is <u>callee</u> of **`main`**

`config_clock` is <u>caller</u> of **`gpio_set_output`**

`gpio_set_output` is <u>callee</u> of **`config_clock`**

Shared use of registers

Only one set of shared registers, must coordinate on which code has "rights" to register during function call

Half of registers designated **callee-owned**

- **Caller** cedes these registers to callee, has no expectation of register contents after call
- **Callee** can freely use these registers

Other half designated **caller-owned**

- **Caller** retains ownership of these registers, register contents must be same after call as before call
- **Callee** can use these registers only if preserve value and restore (thus fulfilling caller's expectation that value is unchanged)

Register ownership

callee-owned `a0-a7,t0-t6`
(a=arg, t=temp)

caller-owned `s0-s11,ra-tp`
(s=saved)

zero	ra	sp	gp
tp	t0	t1	t2
s0/fp	s1	a0	a1
a2	a3	a4	a5
a6	a7	s2	s3
s4	s5	s6	s7
s8	s9	s10	s11
t3	t4	t5	t6

The need for scratch space

Callee has free use of all callee-owned registers, but could need more than that. Why?

- has many local variables with overlapping live ranges

- if callee makes a call to another function, has to cede all callee-owned registers to it

- if callee needs to use a caller-owned register, has to preserve value somewhere so can restore later

Where else can we store additional values and/or copy register values for safekeeping?

The stack to the rescue!

Reserve section of memory to store data for executing function

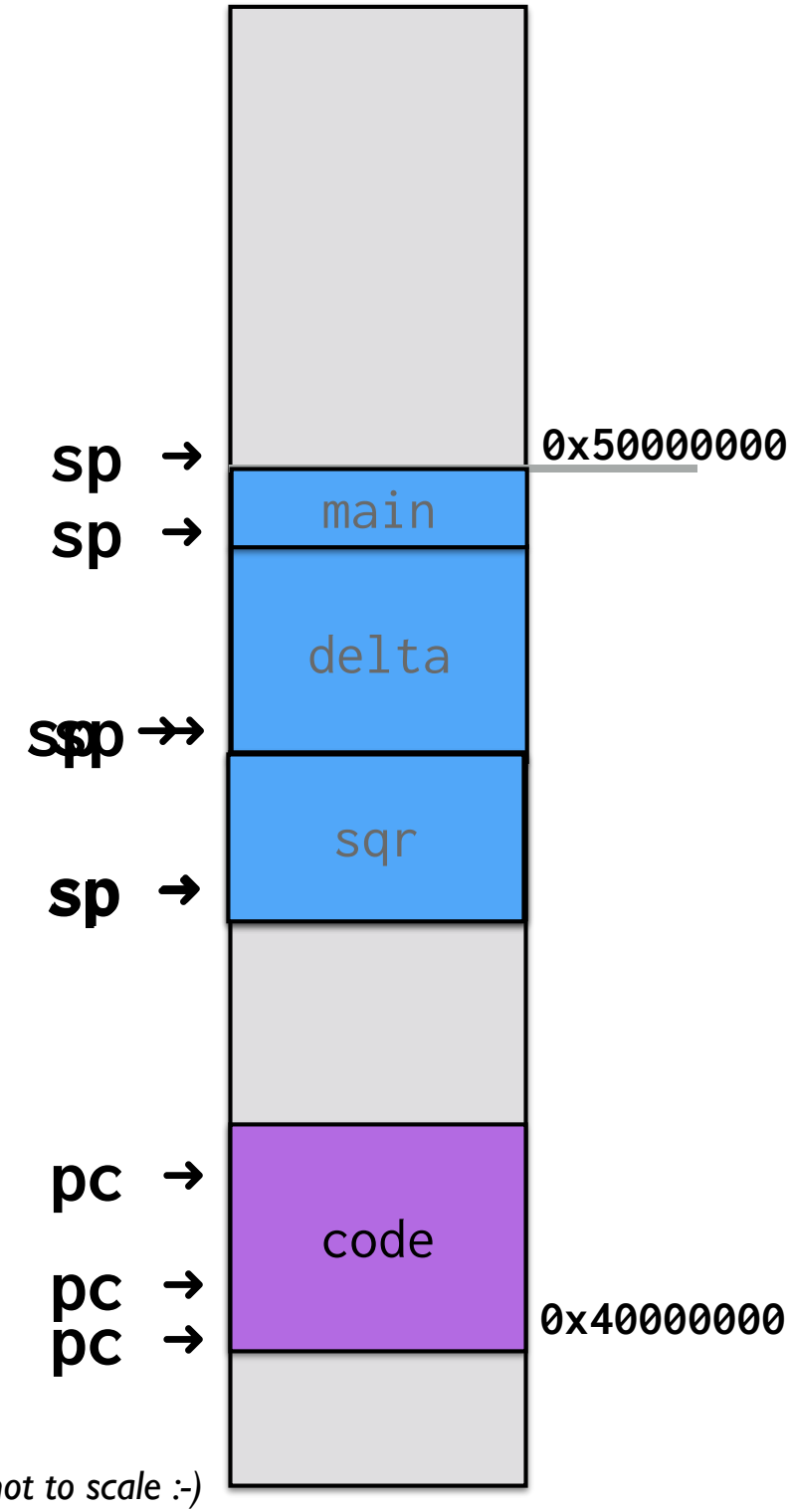
Stack frame allocated per function invocation
Can store local variables, scratch values, saved registers

sp points to location in memory of lastmost value pushed
Per function call:
- Function entry: decrement **sp** makes space for stack frame ("push")
  - During function: access data in frame using **sp**-relative offset
- Function exit: increment **sp** to clean up frame ("pop")

Stack is LIFO, last frame pushed is first frame popped
Stack memory grows down (inner frames at lower addresses)

```
// start.s
lui   sp,0x5000
jal   ra,main
```

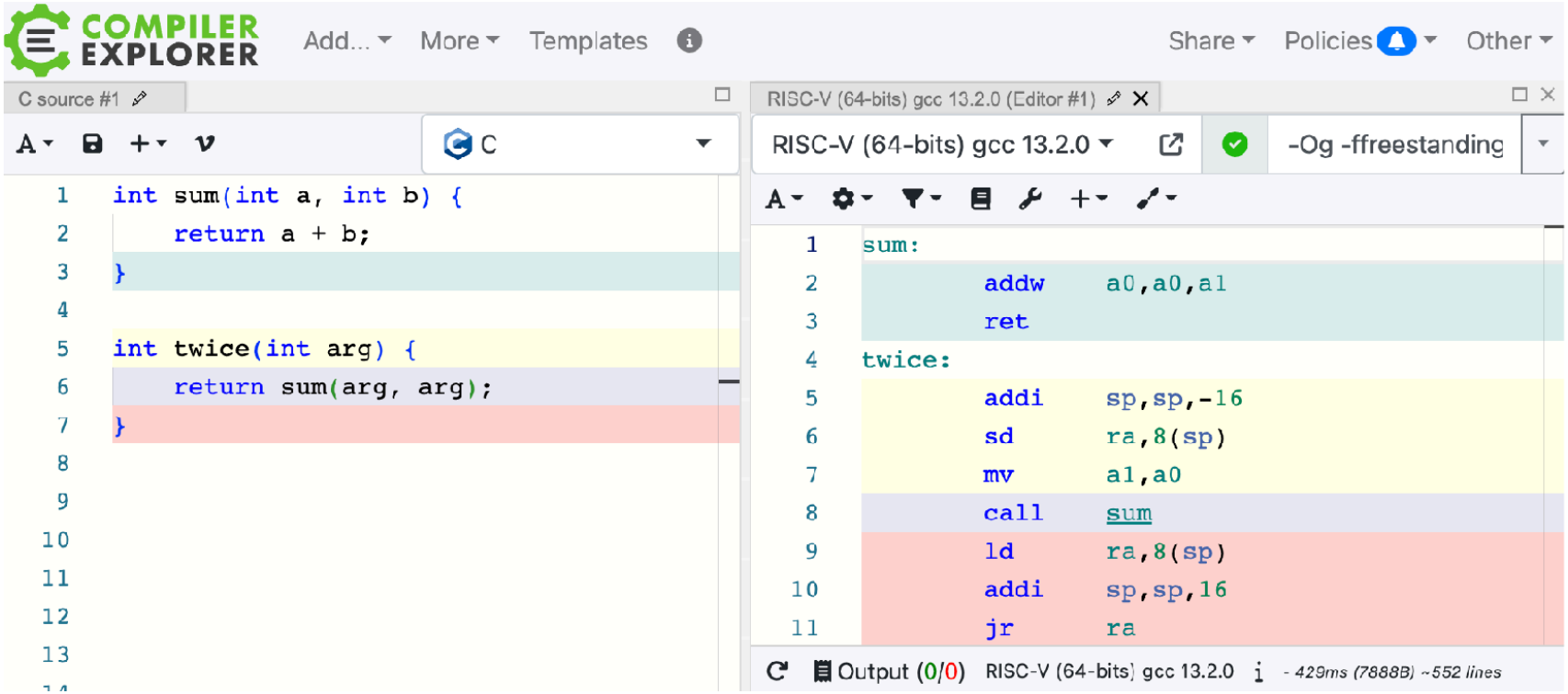
```
void main(void)
{
   delta(3, 7);
}

int delta(int a, int b)
{
  int diff = sqr(a) - sqr(b);
  return diff;
}

int sqr(int v)
{
  return v * v;
}
```



Memory diagram not to scale :-)

Compiler Explorer

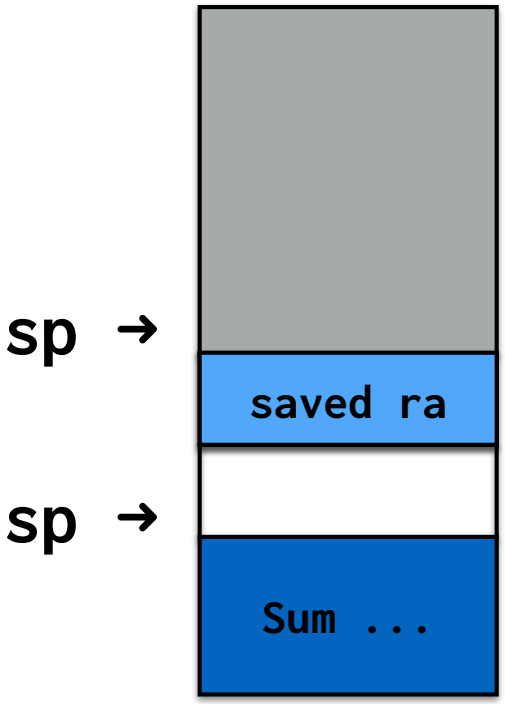
Let's use it to see function-call mechanisms and use of stack



https://godbolt.org/

Stack operation

```
twice:
        addi    sp,sp,-16
        sd      ra,8(sp)
        mv      a1,a0
        call    sum
        ld      ra,8(sp)
        addi    sp,sp,16
        jr      ra
```




```
int twice(int arg) {
    return sum(arg, arg);
}
```

C vs assembly smackdown

Why C?

Variable names, type system

Function decomposition, control flow

Portable abstractions

Consistent semantics

Compiler back-end doing heavy lifting - yay!

Why assembly?

Execution is always in asm, this is the real deal -- WYSIWYG

Ability to drop down and review/debug asm is key

Certain hardware features only accessible via asm

Hand-code in asm for optimization or obtain precise timing

Abstraction FTW!

good

```
    lui     a0,0x2000
    addi    a1,zero,1
    sw      a1,0x30(a0)

loop:
    xori    a1,a1,1
    sw      a1,0x40(a0)

    lui     a2,0x3f00
delay:
    addi    a2,a2,-1
    bne     a2,zero,de

    j       loop
```

better

```
void pause(int count) {
    while (count-- != 0) {}
}

void main(void) {
    volatile unsigned int *PB_CFG0 = 0x2000030;
    volatile unsigned int *PB_DATA = 0x2000040;

    *PB_CFG0 = 1;   // config as output

    while (1) {
        *PB_DATA = 1;   // on
        pause(0x3f00000);
        *PB_DATA = 0;   // off
        pause(0x3f00000);
    }
}
```

best!

```
void main(void) {
    timer_init();
    gpio_init();
    gpio_set_output(pin);

    while (1) {
        gpio_write(pin, 1);
        timer_delay_ms(SECOND);
        gpio_write(pin, 0);
        timer_delay_ms(SECOND);
    }
}
```